BOCCONI UNIVERSITY · COMPUTER VISION & IMAGE PROCESSING

Feedback-Augmented RT-DETR

for Small Object Detection

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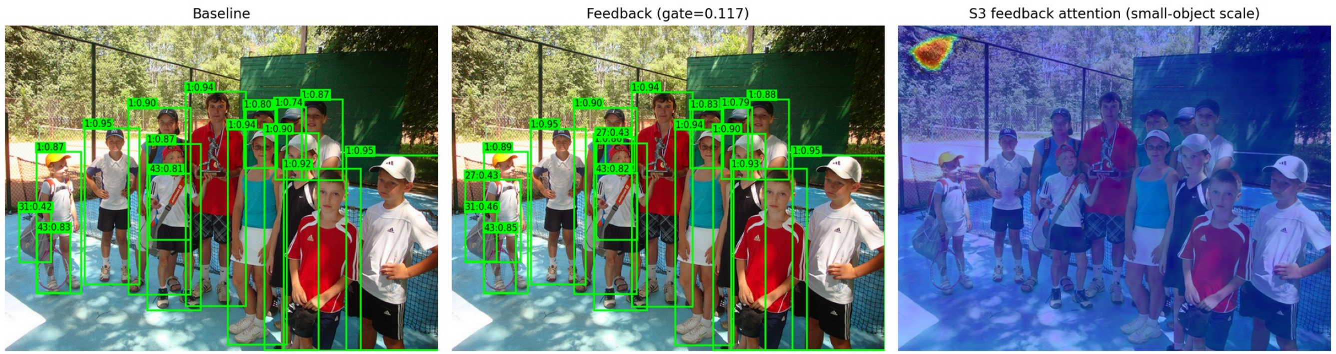
Final Project · April 2026

Small objects are harder to detect



AP_l (large)

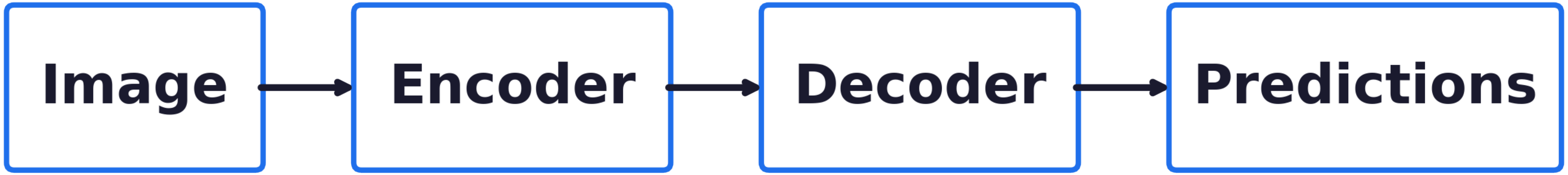
67.7

AP_s (small)

34.7

33-point gap. Why?

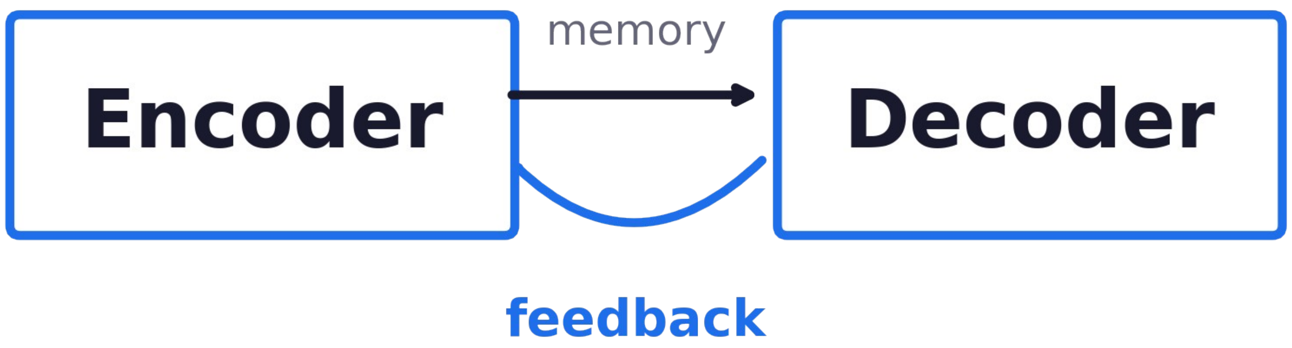
Baseline RT-DETR



one-way information flow

Encoder memory is computed once. The decoder cannot revise it.

Our idea: let the model refine its own features



Feed early decoder predictions back into the encoder memory.

Attention, in one line

Q
what we search for

K
where we look

V
information we extract

$$\mathrm{Attn}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \mathrm{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d}}\right)\mathbf{V}$$

Attention lets the model focus on important parts of the image.

Our feedback rule

m
encoder memory
(features)

g
gate
(how much feedback)

t
decoder output
(predictions)

$$\mathbf{m}' = \mathbf{m} + \mathbf{g} \cdot \mathrm{Attn}(\mathbf{m}, \mathbf{t})$$

We update the encoder using the decoder's own predictions.

First attempt (v1)

Add the feedback module. Train. Toggle it off at inference.

$$\Delta \; AP_S = 0.00$$

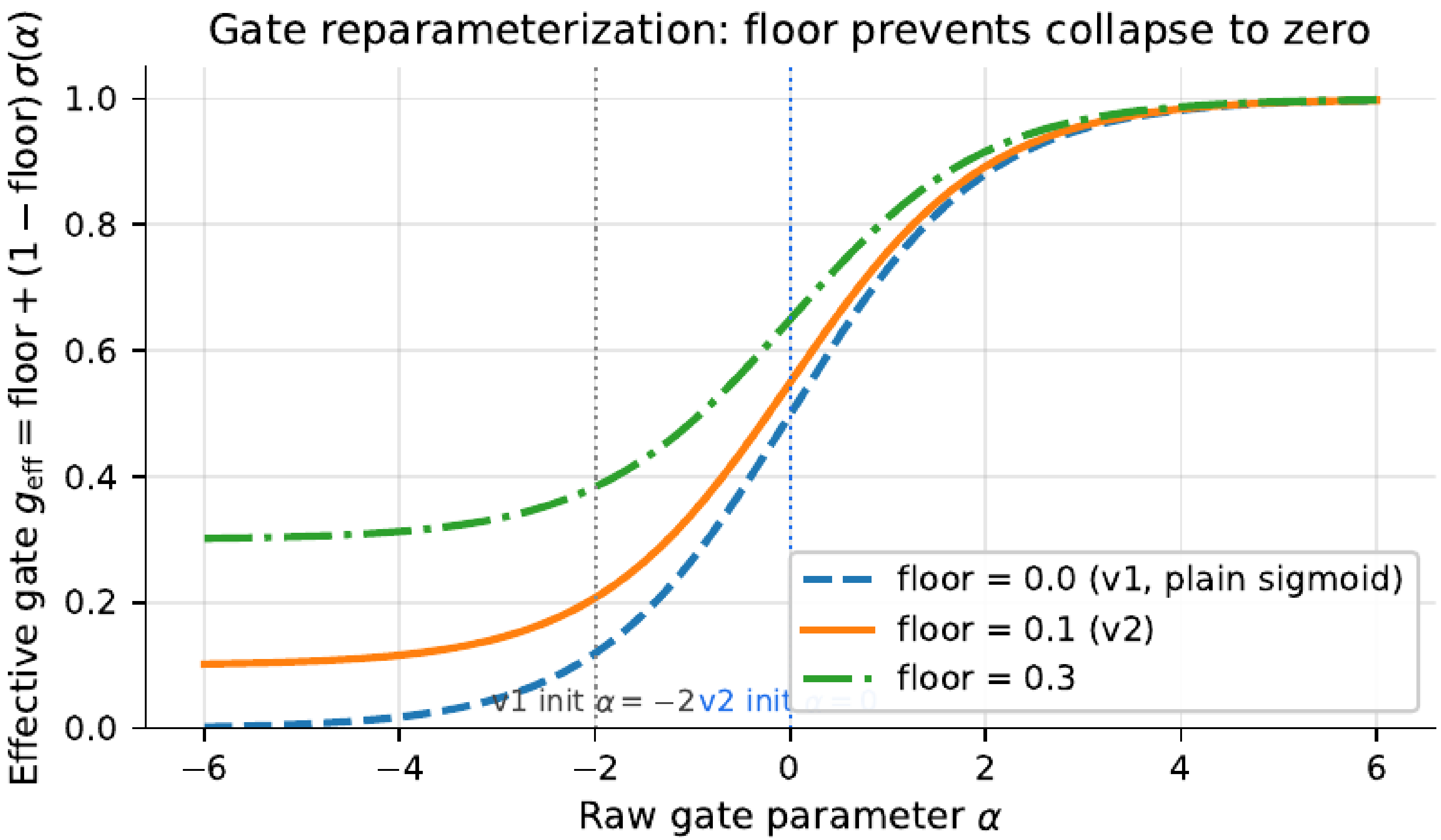
The mechanism contributed nothing.

Why v1 failed: the gate collapsed

Plain sigmoid gate.

α drifts to −∞ → gate ≈ 0

Feedback signal is multiplied by zero before reaching memory.



Two fixes — zero new parameters

1. Gate floor

$$g_{\text{eff}} = 0.1 + 0.9\,\sigma(\alpha)$$

feedback can never be silenced

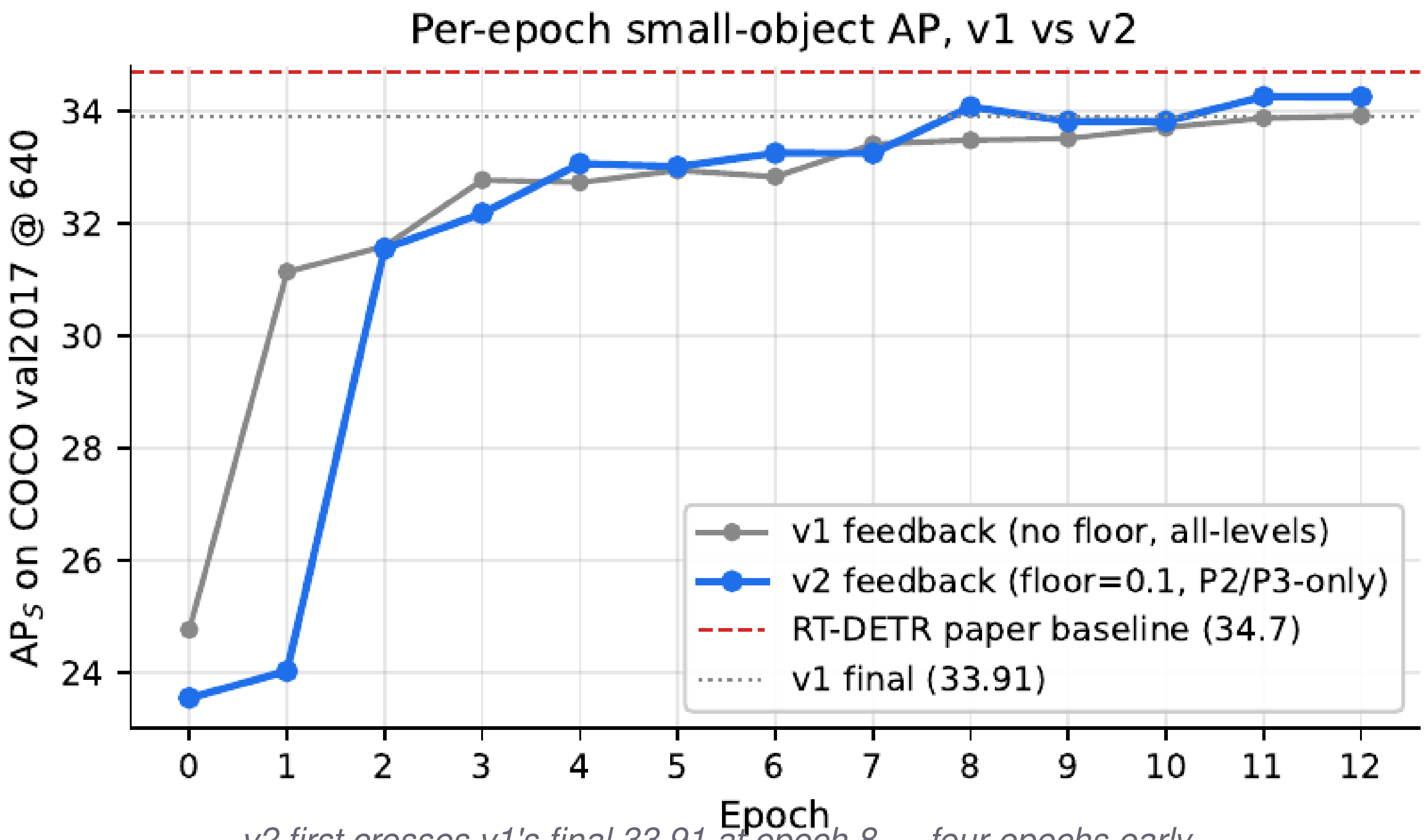
2. P2 / P3 only

refine where small objects live

skip P4, P5 — focus the gradient

Reparameterize the gate. Refine only the levels small objects live in.

Training: v2 climbs faster, ends higher

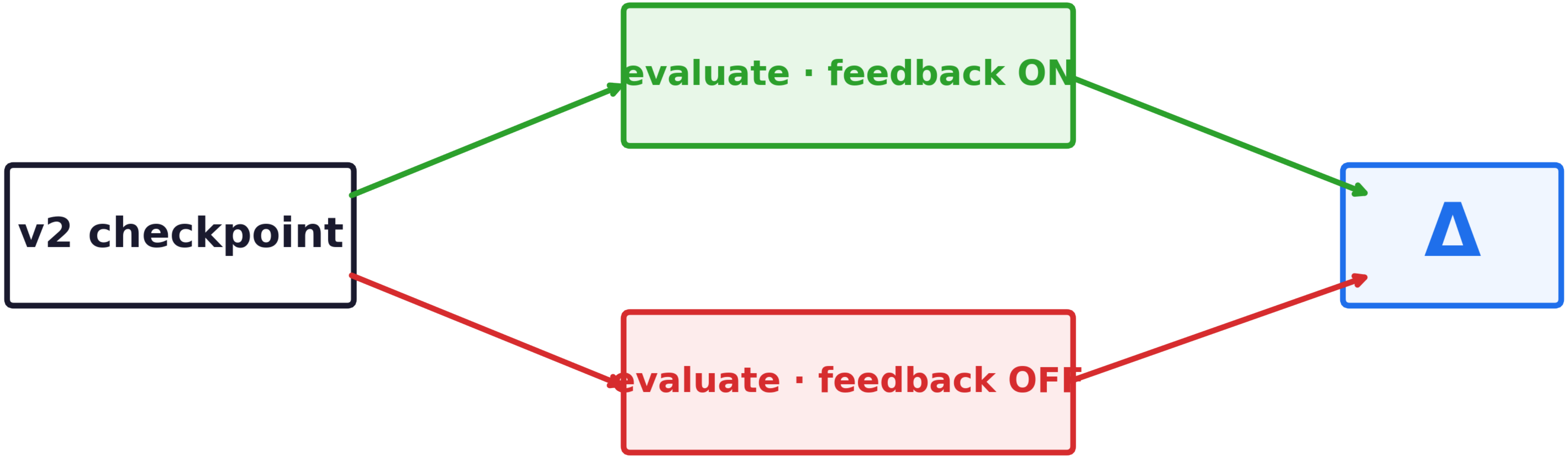


v2 first crosses v1's final 33.91 at epoch 8 — four epochs early.

Final performance (with feedback ON)

v1	v2
33.91	34.30
AP_s	AP_s +0.4

Ablation: same model, only the switch changes



Toggle feedback ON or OFF at inference. Everything else identical.

Causal effect of feedback

ON	OFF	Δ
34.25	**33.26**	**+0.99**
feedback enabled	*feedback bypassed*	*the contribution*

Feedback has a real causal impact.

Higher resolution → bigger gain

640 × 640

34.25

→

800 × 800

37.01

+2.76 from more pixels.

Feedback helps. Spatial sampling helps too.

Trade-offs

+

Better small-object detection

+0.99 AP_S causal contribution

−

Slower inference (FPS drops)

53 FPS → 26 FPS (cost is the P2 level, not feedback)

Conclusion

1. **Feedback improves small-object detection**

   +0.99 AP_s causal contribution at inference.

2. **Only works with proper design**

   Floor the gate; restrict to small-object levels.

3. **Modest but real**

   Consistent across metrics; reproducible from the same checkpoint.

Limitations

- **Single training run**

  no multi-seed variance

- **Below paper baseline**

  13-epoch finetune vs 72-epoch from scratch

- **Increased latency**

  ~2× slower (cost is the P2 level)

Thank you.

Questions?

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github.com/HatemSaadallah/feedback-augmented-rtdetr